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SECTOR & TRADE: ICT & Multimedia, Networking and Internet Technologies
RQF LEVEL 4 | ACADEMIC YEAR 2024-2025 | 3rd December 2025
MODULE TITLE AI INTO ELECTRONIC DEVICES

DURATION 3 Hours
MAXIMUM MARKS 100 Marks
Trainer MPAYIMANA CYIZA Landry
Trainee **……**

INSTRUCTIONS TO STUDENTS

- This paper contains **Section A** and **Section B**.
- **All questions in Section A are compulsory (70 Marks).**
- **Attempt ANY TWO (2) questions in Section B (30 Marks).**
- Use the answer booklet provided.
- Write clearly; unclear answers score zero.

SECTION A (70 Marks) 14 Questions × 5 Marks Each — ALL COMPULSORY

Q1. Which term refers to the sub-field of AI where systems learn patterns automatically from data?
A) Deep Learning
B) Machine Learning
C) Embedded Systems
D) IoT

Q2. Which of the following **best describes an embedded system**?
A) A general-purpose computer
B) A smartphone application
C) A dedicated computing unit designed to perform specific tasks
D) A cloud-based database

Q3. Which file format is MOST commonly used for storing **tabular AI training data**?
A) MP4
B) PNG
C) CSV
D) WAV

(True or False)

Q4. Deep learning uses neural networks with multiple layers to extract complex patterns.
Q5. Data preprocessing includes removing noise, cleaning errors, and normalizing values.
Q6. Unsupervised learning requires labeled datasets.

(Fill-in-the-blank)

Q7. The process of dividing a dataset into **training** and **testing** parts is called _____.
Q8. A complete pass of the entire dataset during model training is known as an _____.
Q9. An optimization algorithm adjusts model _____ to reduce loss.

Q10. Which of the following is an example of **supervised learning**?
A) Clustering vibration patterns
B) Classifying images of “fault” vs “normal”
C) Grouping unlabeled temperature data
D) Robot learning by reward only

Q11. Which library is commonly used to import and analyze AI datasets in Python?
A) Flask
B) NumPy
C) Pandas
D) TensorFlow Serving

Q12. A well-written AI project document includes problem statement, objectives, data plan, hardware requirements, and workflow.

Q13. The Python command used to import a CSV file is:
data = pd._____("file.csv")

Q14. Which of the following prevents **overfitting** in AI model development?
A) Using more epochs
B) Using only training data
C) Splitting dataset into training and testing
D) Ignoring noise in data

SECTION B (30 Marks) Answer ANY TWO Questions Only Each Question = 15 Marks

Q15. a) Define Artificial Intelligence and describe its relevance in electronic systems. (5 marks)
b) Describe the components of a well-prepared AI project document. (5 marks)
c) Provide an example of a technical, social, and economic objective in an AI project. (5 marks)
Q16. a) Explain the different categories of AI data and give an example of each. (5 marks)
b) Describe FOUR data preprocessing techniques and explain their importance. (5 marks)
c) Explain how to correctly create and label a dataset file for model training. (5 marks)
Q17. a) Explain the purpose of splitting data into training and testing sets. (5 marks)
b) Using simple language, describe loss function, epoch, and optimization algorithm. (5 marks)
c) Summarize supervised, unsupervised, and reinforcement learning with examples. (5 marks)
Q18. a) Describe the process of importing an AI dataset into Python using Pandas. (5 marks)
b) Explain how an AI model is trained using algorithms such as Logistic Regression or Neural Networks. (5 marks)
c) Describe THREE techniques used to evaluate AI model performance. (5 marks)

print(“GOOD LUCK!!!!!!!!!!!!!!!”) ;